



USN _____

NMAM INSTITUTE OF TECHNOLOGY, NITTE
(An Autonomous Institution affiliated to VTU, Belagavi)
Third Semester B.E. (ISE) (Credit System) Degree Examinations
Supplementary Examinations – August 2025

21IS302 – DATA STRUCTURES

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.*

		Unit – I	Marks	BT*	CO*	PO*
1.	a)	What is a stack? Explain primitive operations on stack.	8	L*3	2	2
	b)	Explain different functions used in dynamic memory allocation.	8	L2	1	1
	c)	Define data structures. With a neat diagram, explain the classification of data structures.	4	L2	1	1
2.	a)	Write a C program to evaluate a postfix expression.	8	L3	2	2
	b)	Evaluate the given postfix expression: A B C – D * + E \$ F + and assume A=6, B=3, C=2, D=5, E=1 and F=7	8	L3	2	2
	c)	Illustrate array of pointers with the help of C program.	4	L2	1	1
3.	a)	Explain the steps to convert infix expression to postfix expression.	8	L3	2	2
	b)	What is recursion? Write a program to find Fibonacci of a number using recursion.	8	L3	1	2
	c)	How do you declare and initialize a pointer in C?	4	L1	1	1
		Unit – II				
4.	a)	What is a queue? Write a function to perform the following operations in an ordinary queue. i) Insert ii) Delete	10	L3	3	2
	b)	How do you represent Doubly linked list in C. Write functions to i) Insert a node at the front end ii) Delete a node from the rear end.	10	L3	4	2
5.	a)	What are the advantages of circular queues? Write a function to perform the following operations on circular queue. i) Insert ii) Delete	10	L2	3	1
	b)	Write a function to perform the following operations on singly linked list i) insert rear ii) delete front	10	L3	4	2
6.	a)	What is priority queue? Explain the types of priority queue.	10	L2	3	1
	b)	Write a function to perform the following operations on circular singly linked list. i) Insert rear ii) Delete rear	10	L3	4	2
		Unit – III				
7.	a)	What is threaded binary tree? Explain the types of threaded binary tree.	10	L2	5	1
	b)	What is meant by tree traversal? Write C functions to traverse the tree in Inorder, Preorder, Postorder	10	L3	5	2
8.	a)	Define Binary Tree. Explain the representation of a binary tree with a suitable example.	8	L2	5	1
	b)	Construct an expression tree for the following expression $a + (b * c) + d * (e + f)$	6	L3	5	2
	c)	Define the following i) Cycle ii) Complete graph iii) Connected graph	6	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
